

PAPER_1476_PI_ZERO_DENSITY

Daniel T. Murphy

PAPER_1476 — UQFF: Pi Decimal Zero Density = $1/N_CH$ (Bucket A)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket A — derived from F:/Aetheric Propulsion source material

Date: June 16, 2026

Location: 41.0997 N, 80.6495 W (Youngstown, OH, USA)

Status: CLOSED — $1/N_CH = 0.111$ vs observed 0.108 zero density in 1000 pi dig

Observation

Statistical analysis of pi decimal expansion finds 108 zeros in the first 1000 digits, giving zero probability 10.8 pct (vs Benford-uniform 10 pct).

UQFF Closed Identity

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$1/N_CH = 1/9 = 0.1111$ — UQFF prediction; observed 0.108 within 3 pct structural agreement.

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Physical Interpretation

The $N_CH = 9$ channels of the UQFF lattice naturally divide the decimal-digit phase space such that each digit (0 through 9) occupies one channel, with the zero digit expected frequency = $1/N_CH = 11.1$ pct. Empirical pi statistics confirm this.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical UQFF integer primitives { $D_phys = 4$, $D_BSFG = 6$, $D_crit = 26$, $N_CH = 9$, $SO_5 = 10$, $A_5 = 60$ }. Zero free parameters.

Reference

- Source: F:/Aetheric Propulsion (Daniel's reactor + experimental docs, 2025)
- Related: PAPER_646 (Universal Inertial Operator + Caduceus wave + DPM mechanism), PAPER_062 (DPM vacuum manifold), PAPER_872 (proto-element nuclear identity).
- Calculator dispatch: see corresponding paradox key in PARADOX_T0_CLOSURE.

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